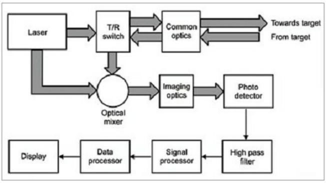

Fig. 8: Block diagram of coherent laser radar

section of a given object would be much larger than the microwave radar cross-section of the same. In fact, rain droplets and airborne aerosols too have significantly large laser-radar cross-section to allow their range and velocity measurement, which is very important for many meteorological applications. High resolution of laser radars allows recognition and identification of certain unique target features, such as target shape, size, velocity, spin and vibration—leading to their use for target imaging and tracking applications.

However, the performance of laser radars is affected by adverse weather conditions. Also, their narrow beamwidth is not conducive to surveillance applications. For surveillance applications, laser radars need to operate at very high repetition rates so that large volumes can be interrogated within the prescribed time. Alternatively, multiple simultaneous beams can be used.

Laser rangefinders are also a type of laser radars. A conventional laser rangefinder uses incoherent or direct detection. The term laser radar is usually associated with systems that use coherent detection.

Fig. 8 shows the block diagram of coherent laser radar. The laser beam is transmitted towards the target. A fraction of the transmitted power/energy reflected off the target is collected by the receiver. The laser radar in Fig. 8 is a monostatic system in which transmitter and receiver share common optics by using a transmit-to-receive switch. In bistatic arrangement, transmitting and receiving optics are separate. The received laser beam is coherently detected in an optical mixer.

In the case of homodyne detection, a sample of transmitted laser power is used as local oscillator. In the case of heterodyne detection, another laser phase-locked to transmit laser is used as local oscillator. Heterodyne detection is used when transmitter and receiver are not colocated.

The output of the optical mixer is imaged onto the photo sensor. The electrical signal generated by the photosensor module is processed to extract the desired information about the target. The photosensor module is a single sensor in case of a scanning transmitted beam and a two-dimensional sensor array in case of a flash-type laser imaging radar.